Linear Classification I: Real Eigenvalues

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Definition: Eigenvalue

Let A be a real 2×2 matrix. A scalar $\lambda \in \mathbb{C}$ is an **eigenvalue** of A if

$$\ker_{\mathbb{C}}(A - \lambda I : \mathbb{C}^2 \to \mathbb{C}^2) \neq \{\mathbf{0}\}.$$

Definition: Eigenvector

Let A be a real 2×2 matrix, and let λ be an eigenvalue of A. A vector $\mathbf{v} \in \mathbb{C}^2 \setminus \{\mathbf{0}\}$ is an **eigenvector associated with** λ if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Definition: Eigendirection

Let A be a real 2×2 matrix. If an eigenvalue λ of A and an associated eigenvector $\mathbf{v}$ are real, the **eigendirection associated with** λ is the line

$$\operatorname{span}\{\mathbf{v}\} = \{c\mathbf{v} : c \in \mathbb{R}\}.$$

Proposition: Eigensolutions

Let $A \in \mathbb{R}^{2\times 2}$. If $A\mathbf{v} = \lambda\mathbf{v}$, where $\lambda \in \mathbb{C}$ and $\mathbf{v} \in \mathbb{C}^2 \setminus \{\mathbf{0}\}$, then

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$$

is a complex-valued solution of the complexified system $\dot{\mathbf{x}} = A\mathbf{x}$, meaning the same differential equation interpreted on $\mathbb{C}^2$.

Proof.

$$\dot{\mathbf{x}}(t) = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}A\mathbf{v} = A\mathbf{x}(t).$$

Theorem: General solution for two distinct real eigenvalues

Let $A \in \mathbb{R}^{2\times 2}$ have distinct real eigenvalues λ_1, λ_2 with associated real eigenvectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^2$. Then $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent, and every real solution of $\dot{\mathbf{x}} = A\mathbf{x}$ has the form

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2, \qquad c_1, c_2 \in \mathbb{R}.$$

Proof. Eigenvectors for distinct eigenvalues are linearly independent, so every initial state has a unique representation

$$\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2.$$

Superposition shows that the displayed function is a solution, and at $t = 0$ it equals $\mathbf{x}_0$. Uniqueness of the linear initial-value problem makes it the only solution.

Theorem: Classification for two distinct real eigenvalues

Let A be a real 2×2 matrix with distinct real eigenvalues λ_1 and λ_2, labeled so that $\lambda_1 < \lambda_2$.

eigenvalue signs	equilibrium type	stability
$\lambda_1 < 0 < \lambda_2$	saddle	unstable
$\lambda_1 < \lambda_2 < 0$	node	asymptotically stable
$0 < \lambda_1 < \lambda_2$	node	unstable
$\lambda_1 < \lambda_2 = 0$	line of equilibria	Lyapunov stable, not attracting
$\lambda_1 = 0 < \lambda_2$	line of equilibria	unstable

Proof. The signs determine whether the factors $e^{\lambda_j t}$ grow or decay in the general solution. If one eigenvalue is zero, its eigendirection is a line of equilibria; the other component decays when its eigenvalue is negative and grows when its eigenvalue is positive.

Proposition: Dominant eigendirection

Let $A \in \mathbb{R}^{2\times 2}$ have distinct real eigenvalues $\lambda_1 < \lambda_2$ with associated real eigenvectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{R}^2$, and let

$$\mathbf{x}(t) = c_1 e^{\lambda_1 t}\mathbf{v}_1 + c_2 e^{\lambda_2 t}\mathbf{v}_2.$$

If $\lambda_1 < \lambda_2 < 0$ and $c_2 \neq 0$, then the trajectory approaches the origin tangent to $\operatorname{span}\{\mathbf{v}_2\}$ as $t \to \infty$. If $0 < \lambda_1 < \lambda_2$ and $c_1 \neq 0$, then the trajectory approaches the origin tangent to $\operatorname{span}\{\mathbf{v}_1\}$ as $t \to -\infty$.

Proof. In the stable case, factoring out the slower exponential gives

$$\mathbf{x}(t) = e^{\lambda_2 t}\left(c_2\mathbf{v}_2 + c_1 e^{(\lambda_1-\lambda_2)t}\mathbf{v}_1\right),$$

and the parenthesized direction tends to $c_2\mathbf{v}_2$. The unstable case is obtained by factoring out $e^{\lambda_1 t}$ and letting $t \to -\infty$.

Corollary: Stable and unstable sets of a linear saddle

Let $A \in \mathbb{R}^{2\times 2}$ have eigenvalues $\lambda_s < 0 < \lambda_u$. Choose real eigenvectors $\mathbf{v}_s, \mathbf{v}_u \in \mathbb{R}^2 \setminus \{\mathbf{0}\}$ such that $A\mathbf{v}_s = \lambda_s \mathbf{v}_s$ and $A\mathbf{v}_u = \lambda_u \mathbf{v}_u$. Then

$$W^s(\mathbf{0}) = \operatorname{span}\{\mathbf{v}_s\}, \qquad W^u(\mathbf{0}) = \operatorname{span}\{\mathbf{v}_u\}.$$

Proof. A solution converges to the origin as $t \to \infty$ exactly when its $\mathbf{v}_u$ coefficient is zero. It converges to the origin as $t \to -\infty$ exactly when its $\mathbf{v}_s$ coefficient is zero.

Question: How do eigenvalues solve an initial-value problem?

Solve

$$\dot{x} = x + y, \qquad \dot{y} = 4x - 2y, \qquad (x(0), y(0)) = (2, -3),$$

and classify the origin.

Solution.

$$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}, \qquad \det(A - \lambda I) = \lambda^2 + \lambda - 6 = (\lambda - 2)(\lambda + 3).$$

Thus

$$\lambda_1 = -3, \qquad \lambda_2 = 2.$$

Solving $(A - \lambda I)\mathbf{v} = \mathbf{0}$ gives

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}, \qquad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Hence

$$\begin{pmatrix} x(t) \\ y(t) \end{pmatrix} = c_1 e^{-3t} \begin{pmatrix} 1 \\ -4 \end{pmatrix} + c_2 e^{2t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The initial condition requires

$$c_1 + c_2 = 2, \qquad -4c_1 + c_2 = -3,$$

so $c_1 = c_2 = 1$. Therefore

$$x(t) = e^{2t} + e^{-3t}, \qquad y(t) = e^{2t} - 4e^{-3t}.$$

The origin is a saddle with

$$W^s(\mathbf{0}) = \operatorname{span}\{(1, -4)\}, \qquad W^u(\mathbf{0}) = \operatorname{span}\{(1, 1)\}.$$

Visualization: Real-eigenvalue geometry

The following table describes trajectories whose coefficients along both displayed eigendirections are nonzero.

portrait	$t \to \infty$	$t \to -\infty$
$\lambda_1 < \lambda_2 < 0$	tangent to $\mathbf{v}_2$	parallel to $\mathbf{v}_1$
$0 < \lambda_1 < \lambda_2$	parallel to $\mathbf{v}_2$	tangent to $\mathbf{v}_1$
$\lambda_s < 0 < \lambda_u$	parallel to $\mathbf{v}_u$	parallel to $\mathbf{v}_s$

The table does not apply to the eigendirection trajectories: each eigendirection is itself an invariant straight line, and a solution that starts on one remains on it. A saddle trajectory with nonzero coefficients along both eigendirections approaches the unstable eigendirection as $t \to \infty$; it does not approach the stable manifold in forward time.

Theorem: Repeated eigenvalue with two eigendirections

Suppose the characteristic polynomial of a real 2×2 matrix A is $(s-\lambda)^2$ and the eigenspace for λ contains two linearly independent eigenvectors. Then

$$A = \lambda I.$$

If $\lambda < 0$, the origin is an asymptotically stable star node; if $\lambda > 0$, it is an unstable star node; if $\lambda = 0$, every point is an equilibrium.

Proof. Every vector $\mathbf{x}$ is a linear combination of the two eigenvectors, so $A\mathbf{x} = \lambda \mathbf{x}$ for every $\mathbf{x}$. Hence $A = \lambda I$ and

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{x}_0.$$

The classification follows.

Definition: Defective eigenvalue

Let A be a real 2×2 matrix. A repeated eigenvalue λ of A is **defective** if

$$\dim\ker(A - \lambda I) = 1.$$

Definition: Generalized eigenvector

Let $A \in \mathbb{C}^{n\times n}$, let $\lambda \in \mathbb{C}$ be an eigenvalue, and let $\mathbf{v} \in \mathbb{C}^n \setminus \{\mathbf{0}\}$ satisfy

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

A vector $\mathbf{w} \in \mathbb{C}^n$ satisfying

$$(A - \lambda I)\mathbf{w} = \mathbf{v}$$

is a **generalized eigenvector associated with $\mathbf{v}$**. The pair $(\mathbf{v}, \mathbf{w})$ is a **length-two Jordan chain**.

Theorem: Solution from a length-two Jordan chain

Let A be a real 2×2 matrix and let $\lambda \in \mathbb{R}$. Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^2$, $\mathbf{v} \neq \mathbf{0}$, and

$$A\mathbf{v} = \lambda\mathbf{v}, \qquad (A - \lambda I)\mathbf{w} = \mathbf{v}.$$

Then $\mathbf{v}$ and $\mathbf{w}$ are linearly independent, and every real solution of $\dot{\mathbf{x}} = A\mathbf{x}$ is

$$\mathbf{x}(t) = e^{\lambda t}\left[(c_1 + c_2 t)\mathbf{v} + c_2\mathbf{w}\right], \qquad c_1, c_2 \in \mathbb{R}.$$

Proof. The two functions

$$e^{\lambda t}\mathbf{v}, \qquad e^{\lambda t}(t\mathbf{v} + \mathbf{w})$$

solve the system. For the second,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left[e^{\lambda t}(t\mathbf{v} + \mathbf{w})\right] = e^{\lambda t}\left[(\lambda t + 1)\mathbf{v} + \lambda\mathbf{w}\right],$$

while

$$Ae^{\lambda t}(t\mathbf{v} + \mathbf{w}) = e^{\lambda t}\left[\lambda t\mathbf{v} + A\mathbf{w}\right] = e^{\lambda t}\left[(\lambda t + 1)\mathbf{v} + \lambda\mathbf{w}\right].$$

If $\mathbf{w}$ were a multiple of $\mathbf{v}$, then $(A - \lambda I)\mathbf{w} = \mathbf{0}$, contradicting $\mathbf{v} \neq \mathbf{0}$. Thus $\mathbf{v}, \mathbf{w}$ are independent, and the two displayed solutions span the two-dimensional real solution space.

Question: What does a defective node look like?

Let

$$A = \begin{pmatrix} \lambda & b \\ 0 & \lambda \end{pmatrix}, \qquad \lambda \in \mathbb{R}, \qquad b \in \mathbb{R} \setminus \{0\}.$$

Find the general solution and classify the origin.

Solution. The eigenspace is

$$\ker(A - \lambda I) = \operatorname{span}\{(1, 0)\}.$$

The system is

$$\dot{x} = \lambda x + by, \qquad \dot{y} = \lambda y.$$

Thus

$$y(t) = y_0 e^{\lambda t}, \qquad x(t) = e^{\lambda t}(x_0 + by_0 t).$$

If $\lambda < 0$, the origin is an asymptotically stable degenerate node; if $\lambda > 0$, it is an unstable degenerate node. In both time directions, generic trajectories become parallel to the unique eigendirection. If $\lambda = 0$, every point on $y = 0$ is an equilibrium and every solution with $y_0 \neq 0$ has $x(t) = x_0 + by_0 t$. Thus the origin belongs to a nonisolated line of equilibria and is unstable.

Question: Can a repeated eigenvalue alone determine the portrait?

Classify

$$A_1 = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}, \qquad A_2 = \begin{pmatrix} -2 & 1 \\ 0 & -2 \end{pmatrix}.$$

Solution. Both matrices have the repeated eigenvalue -2. For A_1, every nonzero vector is an eigenvector, so the origin is an asymptotically stable star node. For A_2,

$$\ker(A_2 + 2I) = \operatorname{span}\{(1, 0)\},$$

so the origin is an asymptotically stable degenerate node. The eigenvalue list does not distinguish these portraits; the eigenspace dimension does.